

Integration

Week 5A

Overview

Definite Integral

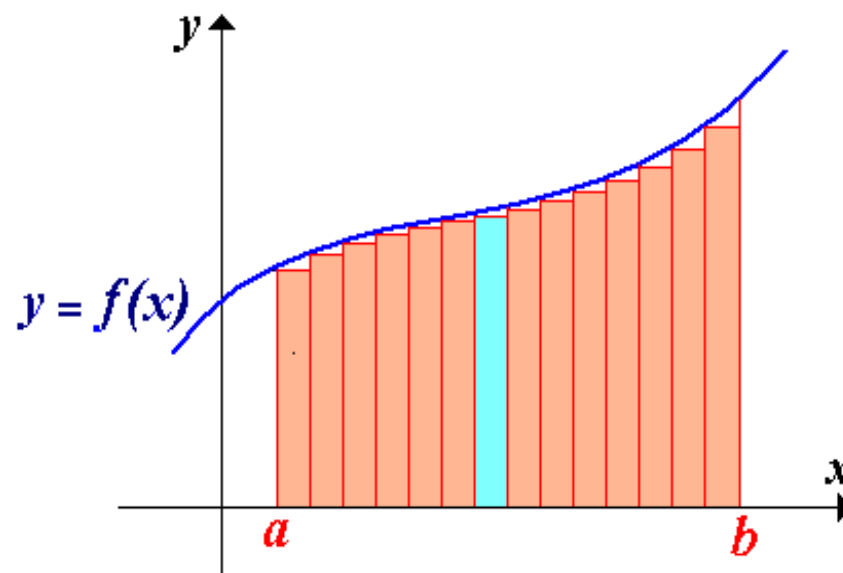
- Key Ideas*
- *Introduction to Fundamental Theorem of Arithmetic*
 - *Properties of Definite Integrals*
 - *Definite Integrals of Power of x*

Definite Integrals and Area

- Key Ideas*
- *Introduction to Area*
 - *The Area under a Curve*

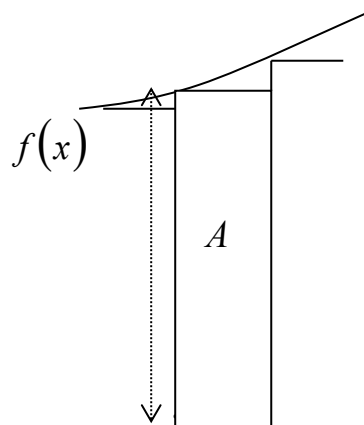
Key Idea – History of the Integration – OPTIONAL

In the seventeenth century, Leibniz and Newton linked the area under a curve to the primitive function.



The area under the curve from $x = a$ to $x = b$ is approximately equal to the sum of all the rectangles.

A typical rectangle of width Δx is enlarged below.



Area of this rectangle $A = f(x) \times \Delta x$

Area under the curve from $x = a$ to $x = b$

$$\approx \sum_{x=a}^{x=b} \text{areas of all the rectangles}$$
$$\approx \sum_{x=a}^{x=b} f(x) \Delta x$$

As $\Delta x \rightarrow 0$, the number of rectangles increases, and the width of the rectangles becomes narrower. Also, the top of the series of rectangles becomes less ‘jagged’ and smoother, gradually approaching the shape of the curve under which it sits. This means the series of rectangles will eventually match *exactly* the area under the curve.

Therefore, to find the exact area, we take the limit as $\Delta x \rightarrow 0$, $A = \lim_{\Delta x \rightarrow 0} \sum_{x=a}^{x=b} f(x) \Delta x$.

In integral calculus, $A = \int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{x=a}^{x=b} f(x) \Delta x$

$\int_a^b f(x) dx$ represents the area under the curve $y = f(x)$ for the range of x -values from $x = a$ to $x = b$.

This is called the integral of $f(x)$ between $x = a$ and $x = b$.

The $\int$ symbol is an old form of S (standing for ‘sum’) and the word ‘integrate’ means to find the limit of the total sum.

Key Idea - The Definite Integral

The **definite integral** is a **number** and is specified in the following way:

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

This is known as the Fundamental Theorem of Calculus.

Where **$F(x)$ is the antiderivative** (or primitive function). The number “ **a** ” is the **lower limit** and the number “ **b** ” is the **upper limit** of integration.

Example: Evaluate the following definite integrals: $\int_3^4 2x + 1 \, dx$.

$$\begin{aligned}\int_3^4 2x + 1 \, dx &= \left[\frac{2x^2}{2} + x \right]_3^4 \\ &= [x^2 + x]_3^4 \\ &= (4^2 + 4) - (3^2 + 3) \\ &= 20 - 12 = 8\end{aligned}$$

Exercise:

Evaluate the following definite integrals:

(a) $\int_{-1}^1 x^2(1-4x) dx$ *expand* $= \int_{-1}^1 (x^2 - 4x^3) dx = \left[\frac{x^3}{3} - \frac{4x^4}{4} \right]_{-1}^1$
 $= \left[\left(\frac{1^3}{3} - 1^4 \right) - \left(\frac{(-1)^3}{3} - (-1)^4 \right) \right]$
 $= \left[\left(\frac{1}{3} - 1 \right) - \left(-\frac{1}{3} - 1 \right) \right] = \frac{2}{3}$

(b) $\int_{-1}^{\frac{1}{2}} (-x^3) dx$
 $= \left[-\frac{x^4}{4} \right]_{-1}^{\frac{1}{2}}$ OR $-\frac{1}{4} [x^4]_{-1}^{\frac{1}{2}}$ *easier*
 $= \left(-\frac{(\frac{1}{2})^4}{4} \right) - \left(-\frac{(-1)^4}{4} \right)$
 $= \left(-\frac{\frac{1}{16}}{4} \right) - \left(-\frac{1}{4} \right) = -\frac{1}{64} + \frac{1}{4} = \frac{15}{64}$

Key Ideas – Properties of Definite Integrals

Some simple properties of definite integrals can be derived from the basic definition, or from the Fundamental Theorem of the Calculus.

1. This property removes the restriction that the lower limit of the integral must be smaller than the upper limit. Interchanging the limits of integration reverses the sign of the integral.

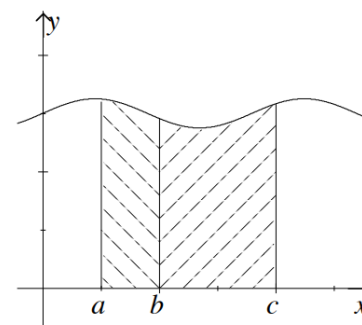
$$\int_b^a f(x) dx = -\int_a^b f(x) dx$$

2. If the upper and lower limits of the integral are the same, the integral is zero. This becomes obvious if we have a positive function and can interpret the integral in terms of ‘the area under a curve’.

$$\int_a^a f(x) dx = 0$$

3. If $a < c < b$, then $\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$

This property states that the integral of a function over two adjacent intervals is equal to the sum of the integrals over each interval. The diagram opposite helps to make this clear if $f(x)$ is a positive function.



Example: Consider $\int_3^5 2x \, dx = \left[x^2 \right]_3^5 = (5^2 - 3^2) = 16$

1. $\int_b^a f(x) \, dx = -\int_a^b f(x) \, dx$

$$\int_5^3 2x \, dx = \left[x^2 \right]_5^3 = (3^2 - 5^2) = 9 - 25 = -16$$

2. $\int_a^a f(x) \, dx = 0$

$$\int_3^3 2x \, dx = \left[x^2 \right]_3^3 = (3^2 - 3^2) = 0$$

3. $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$

$$\begin{aligned} \int_3^5 2x \, dx &= \int_3^4 2x \, dx + \int_4^5 2x \, dx = \left[x^2 \right]_3^4 + \left[x^2 \right]_4^5 \\ &= ((4^2 - 3^2) + (5^2 - 4^2)) = 16 \end{aligned}$$

Exercise:

Show that $\int_0^3 x^2 dx = -\int_3^0 x^2 dx$

$$\begin{aligned} \text{LHS} &= \int_0^3 x^2 dx = \left[\frac{x^3}{3} \right]_0^3 \\ &= \left[\frac{3^3}{3} - \frac{0^3}{3} \right] = [9 - 0] \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= -\int_3^0 x^2 dx = -\left[\frac{x^3}{3} \right]_3^0 \\ &= -\left[\frac{0^3}{3} - \frac{3^3}{3} \right] \end{aligned}$$

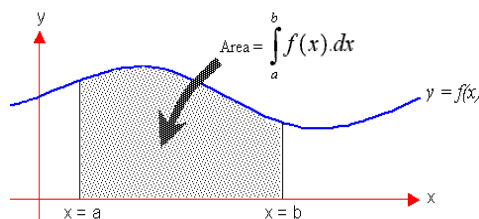
$\therefore \text{LHS} = \text{RHS}$

$$= -[0 - 9] = -(-9) = 9$$

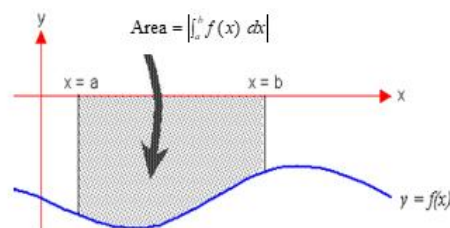
Key Idea – Introduction to Area

We have defined the definite integral as $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$ without **placing any restrictions on whether the function $f(x)$ is positive or negative.**

- If $f(x)$ is positive between a and b , $\int_a^b f(x) dx$ is equal to the **area between the curve $y = f(x)$, the x -axis and the ordinates $x = a$ and $x = b$, which we refer to as ‘the area under the curve’.**



- If $f(x)$ is negative between a and b , then $\int_a^b f(x) dx$ will also be negative. Now areas are, by definition, positive. So, to find the area, if $f(x)$ is negative we take the **absolute value** of $\int_a^b f(x) dx$, i.e. $\left| \int_a^b f(x) dx \right|$.



This makes it **very clear that a definite integral does not always** represent the area under a curve. We have found that

1. If $f(x)$ is positive between a and b , then $\int_a^b f(x) dx$ does represent the area under the curve.
2. If $f(x)$ is negative between a and b , then $\left| \int_a^b f(x) dx \right|$ represent the area above the curve, since the value of $\int_a^b f(x) dx$ is negative.
3. If $f(x)$ is sometimes positive and sometimes negative between a and b , then $\int_a^b f(x) dx$ measures the difference in area between the part above the x -axis and the part below the x -axis.

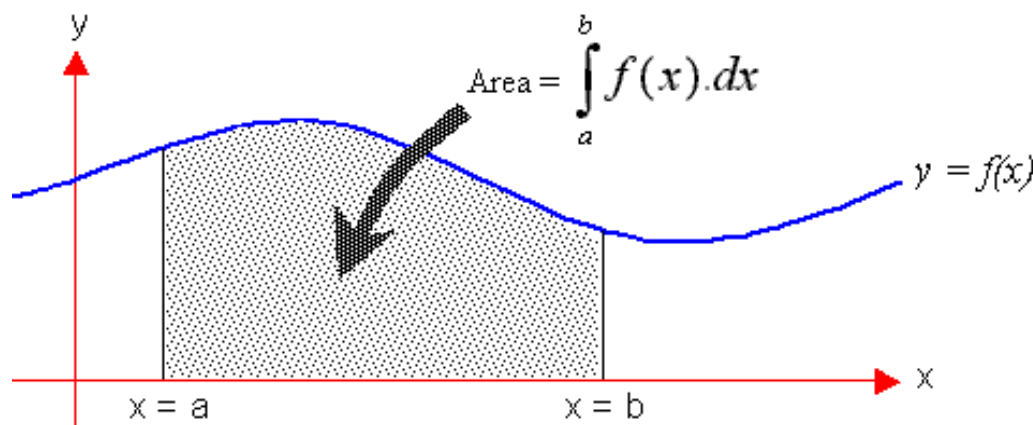
We will look at each of these in more detail below.

WARNING:

In working out area problems you should always sketch the curve first.

Key Idea – Area between the Curve and the x -axis (Area under the Curve)

The definite integral of a function gives the signed area between the curve $y = f(x)$, the x -axis, and the vertical lines $x = a$ and $x = b$. Areas above the x -axis are positive.



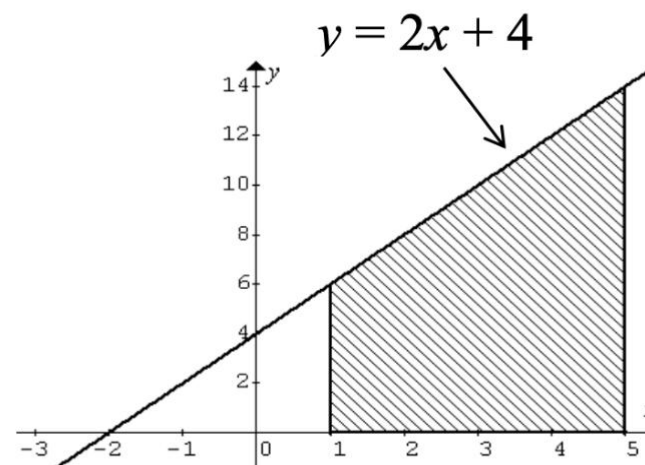
$$\text{Area} = \int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

Note: As you are calculating an area, your answer must include units².

Example:

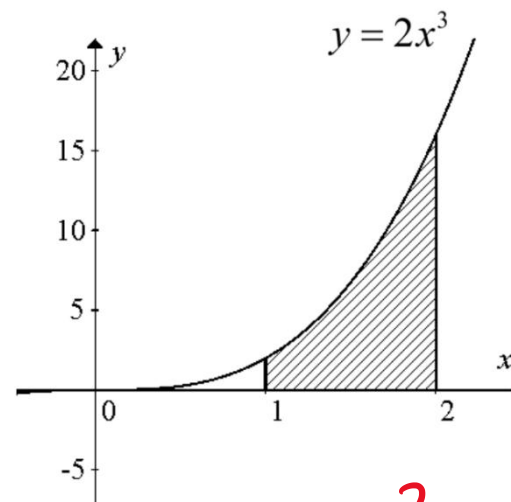
Calculate the area of the shaded region in the graph below.

$$\begin{aligned} A &= \int_1^5 (2x + 4) dx = \left[x^2 + 4x \right]_1^5 \\ &= (5^2 + 20) - (1^2 + 4) \\ &= (45) - (5) \\ &= 40 \text{ units}^2 \end{aligned}$$



Exercise:

Calculate the area of the shaded region in the graph below.



$$A = \int_1^2 2x^3 dx$$

$$= 2 \int_1^2 x^3 dx$$

$$= 2 \left[\frac{x^4}{4} \right]_1^2$$

$$= 2 \left[\frac{2^4}{4} - \frac{1^4}{4} \right]$$

$$= \frac{15}{2} \text{ u2}$$



$$\text{OR } \left[\frac{2x^4}{4} \right]_1^2$$

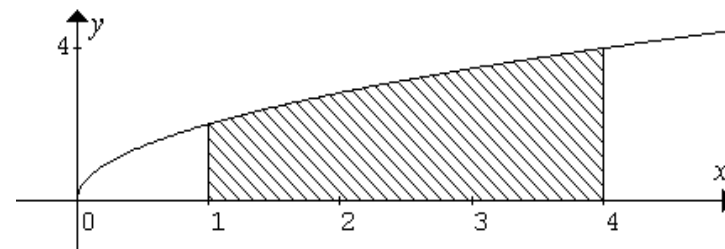
$$= \left[\frac{x^4}{2} \right]_1^2$$

$$= \left(\frac{2^4}{2} \right) - \left(\frac{1^4}{2} \right)$$

$$= 15/2 \text{ u2}$$

Exercise:

- (a) Write the definite integral expression required to calculate the shaded area under function $y = 2\sqrt{x}$.
- (b) Find the area of under function $y = 2\sqrt{x}$.



$$(a) \quad A = \int_1^4 2\sqrt{x} \, dx = 2 \int_1^4 \sqrt{x}$$

$$(b) \quad = 2 \int_1^4 x^{\frac{1}{2}} \, dx$$

$$= 2 \left[\frac{2x^{3/2}}{3/2} \right]_1^4 = 2 \left[\frac{2x^{3/2}}{3} \right]_1^4$$

$$= 2 \left[\left(\frac{2 \times 4^{3/2}}{3} \right) - \left(\frac{2 \times 1^{3/2}}{3} \right) \right]$$

$$= \frac{28}{3} \, u^2$$

Exercise:

Write an integral which can be used to calculate the area enclosed by $y = 6x - x^2$ and the x-axis.

To find the integral you first need to find intercepts and it's helpful to draw so that you know if the area enclosed is above or below the x-axis.

• $y = 6x - x^2$ $a = -1 < 0 \therefore$ concave down $\cap$

• y-int, let $x = 0$: $y = 6 \times 0 - 0^2 = 0 \therefore (0, 0)$

x-int, let $y = 0$: $6x - x^2 = 0 \Rightarrow x(6 - x) = 0$

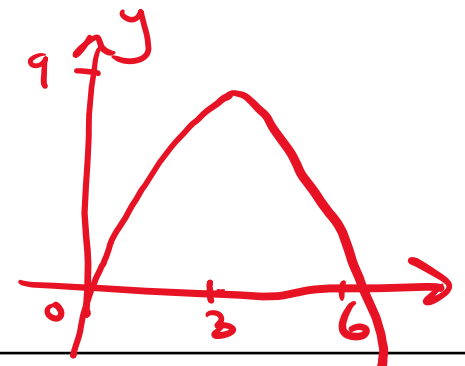
$\Rightarrow x = 0, 6 - x = 0$
 $x = 0 \quad x = 6$

$$A = \int_0^6 (6x - x^2) dx$$

$$= \left[\frac{6x^2}{2} - \frac{x^3}{3} \right]_0^6 = \left[3x^2 - \frac{x^3}{3} \right]_0^6$$

$$= \left[\left(3 \times 6^2 - \frac{6^3}{3} \right) - \left(3 \times 0^2 - \frac{0^3}{3} \right) \right]$$

$$= 36 \text{ u}^2$$

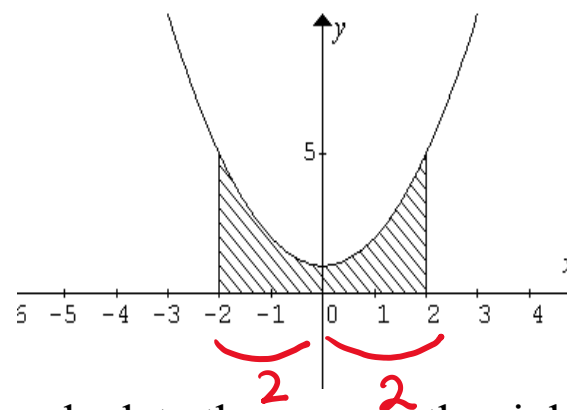


Symmetrical Graphs.

When evaluating definite integrals, calculations are often simpler when one of the limits is zero. For graphs that are symmetrical about the y -axis, the area on the left side of the y -axis is equal to the area on the right side. Instead of calculating both regions separately, calculate the area on one side of the y -axis and then double the result.

Example:

Calculate the shaded area shown in graph of $y = x^2 + 1$



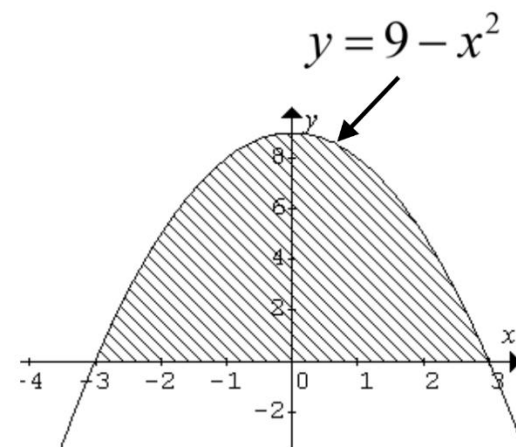
Because the graph is symmetrical about the y -axis, we can calculate the area on the right-hand side of the y -axis and then double it to find the total area

$$\text{Area} = \int_{-2}^2 x^2 + 1 \, dx = 2 \int_0^2 x^2 + 1 \, dx$$

Exercise:

A graph of $y = 9 - x^2$ is drawn below. Calculate the area of the shaded region.

$$\begin{aligned} A &= \int_{-3}^3 (9 - x^2) dx = 2 \int_0^3 (9 - x^2) dx \\ &= 2 \left[9x - \frac{x^3}{3} \right]_0^3 \\ &= 2 \left[\left(9 \times 3 - \frac{3^3}{3} \right) - \left(9 \times 0 - \frac{0^3}{3} \right) \right] \\ &= 2 \left[(27 - 9) - 0 \right] \\ &= 36 \text{ u}^2 \end{aligned}$$



Note: if the graph was not given you would have to find the x -intercept first.